

Kwun Yip, Fung (Samuel)

Postdoctoral Research Associate in Atmospheric and Oceanic Science in Princeton University;
ORCID: 0000-0003-1213-165X Email: kf8430@princeton.edu

Education

The University of Texas at Austin

Texas, US

Ph.D., Geological Sciences, GPA: 3.99/4.00

2018 – 2023

Dissertation: “Improved urban extreme weather simulation by capturing urban heterogeneity”

The Chinese University of Hong Kong

Hong Kong

M.Phil., Earth and Atmospheric Sciences, GPA 3.71/4.00

2016 – 2018

Thesis: “Comparing the anthropogenic heat and global warming impacts on extreme rainfall in Southern China Pearl River Delta based on dynamical downscaling”

The Chinese University of Hong Kong

Hong Kong

B.Sc., Earth System Science, First Class Honours, GPA 3.39/4.00

2012 – 2016

Thesis: “The importance of aerosol-phase radical-radical reactions on the heterogeneous OH oxidation of organic aerosols”

Funded Project

- FI, Effect of urban resilience enhancing strategies on modulating extreme rainfall in Houston, Future Investigators in NASA Earth and Space Science and Technology (FINESST) fellowship, NASA, 09/01/2020 – 08/30/2023, \$134,976

Publication

- Fung, K. Y., Zhu, P., Zhang, X., Zhang, J. A., Wang, C. K., & Liu, B. (2026). Evaluating the Impact of a Scale-Aware Three-Dimensional Turbulent Kinetic Energy Scheme on Tropical Cyclone Forecasts in the Hurricane Analysis and Forecasting System. *Tropical Cyclone Research and Review*. doi: 10.1016/j.tccr.2026.02.001
- Kamath, H. G., Lin, T. S., Sudharsan, N., Fung, K. Y., Zonato, A., Singh, M., ... & Niyogi, D. (2026). Urban heat mitigation under realistic roof and land area constraints. *Environmental Research Letters*, 21(2), 024001. doi: 10.1088/1748-9326/ae32a7
- Kamath, H. G., Lin, T.-S., Sudarshan, N., Fung, K. Y., Zonato, A., Singh, M., Yang, Z.-L., Coudert, M., Niyogi, D. Urban Heat Mitigation Under Realistic Roof and Land Area Constraints. (Submitted to *Environmental Research Letters*)
- Zhu, P., Fung, K. Y., Zhang, X., Zhang, J. A., Bao, J. W., Wang, C. K., ... & Wang, W. (2025). Toward a unified parameterization of three dimensional turbulent transport in high resolution numerical weather prediction models. *npj Climate and Atmospheric Science*, 8(1), 1-13. doi: 10.1038/s41612-025-01117-6
- Fung, K. Y., Yang, Z.-L., Niyogi, D. (2024). Capturing urban heterogeneity enhances tropical cyclones simulation in Houston. *Journal of Geophysical Research: Atmospheres*, 129(19). doi: 10.1029/2024JD040807
- Fung, K. Y., Yang, Z.-L., Martilli, A., Krayenhoff, E. S., Niyogi, D. (2024). Prioritizing equity in urban heat mitigation. *PNAS nexus*, 3(9). doi: 10.1093/pnasnexus/pgae360
- Fung K. Y., Yang, Z.-L., & Niyogi, D. (2022). Improving the local climate zone classification with building height, imperviousness, and machine learning for urban models. *Computational Urban Science*, 2(1). doi: 10.1007/s43762-022-00046-x
- Hu, C., Tam, C., Li, X., Huang, K., Ren, C., Fung, K. Y., & Wang, Z. (2023). Mega-city Development Impact on hourly extreme rainfall over the South China Greater Bay Area under near-future climate warming. *Urban Climate*, 48, 101389. doi:10.1016/j.uclim.2022.101389
- Fung, K. Y., Tam, C.-Y., Lee, T. C., & Wang, Z. (2021). Comparing the Influence of Global Warming and Urban Anthropogenic Heat on Extreme Precipitation in Urbanized Pearl River Delta Area Based on Dynamical Downscaling. *Journal of Geophysical Research: Atmospheres*, 126(21). doi: 10.1029/2021JD035047
- Hu, C., Fung, K. Y., Tam, C.-Y., & Wang, Z. (2021). Urbanization Impacts on Pearl River Delta Extreme Rainfall Sensitivity to Land Cover Change Versus Anthropogenic Heat. *Earth And Space Science*, 8(3). doi: 10.1029/2020ea001536

Research Experience

Princeton University | Atmospheric and Oceanic Sciences

New Jersey, US

Postdoctoral Research Associate

Jul 2026 – Now

- Evaluating and improving the System for High-resolution prediction on Earth-to-Local Domains (SHIELD) model

University of Miami | Cooperative Institute for Marine and Atmospheric Studies (CIMAS); Atlantic Oceanographic and Meteorological Laboratory | Hurricane Research Division

Florida, US

Sep 2023 – Jun 2026

Postdoctoral Associate

- Improved the Hurricane Analysis and Forecast System (HAFS) with the development of a scale-aware turbulent kinetic energy planetary boundary layer scheme
- Improved the gust wind model for HAFS
- Analyzed field observations to improve understanding and prediction of hurricane tracks, intensity, and structure

The University of Texas at Austin | Department of Geological Sciences

Texas, US

Ph.D. Student

Sep 2018 – Aug 2023

- Analyzed multiple machine learning algorithms and urban auxiliary datasets in generating Local Climate Zone (LCZ) maps
- Studied the performance of implementing the LCZ classification scheme into the Weather and Research Forecasting model (WRF) in different weather conditions
- Studied the impacts of urban overheating mitigation strategies on social aspects
- Investigated antecedent soil conditions on hurricane rainfall
- Investigated the urbanization impacts of tornadoes

Purdue University | Department of Agronomy

Indiana, US

Summer Visitor

Jun 2019 – Aug 2019

- Generated the LCZ map for the Austin area using SAGA GIS
- Incorporated LCZ schemes into the Noah and Noah-MP land surface models in WRF

The Chinese University of Hong Kong | Earth and Atmospheric Sciences

Hong Kong

Master's Student

Aug 2016 – Aug 2018

- Used urban-WRF to downscale a global circulation model (GFDL-ESM2M)
- Analyzed the mechanisms (e.g. 2m temperature, CAPE, circulation)
- Presented results in various international conferences (AGU, AOGS – EGU)

The Chinese University of Hong Kong | Earth System Science Programme

Hong Kong

Undergraduate Researcher

Sep 2015 – Jul 2016

- Improved a kinetic molecular model (KMGAP) by importing radical chemistry
- Captured realistic physical and chemical properties change of aerosols during heterogeneous OH oxidation

Hong Kong Observatory

Hong Kong

Research Intern in Development Division

Jun 2015 – Jul 2015

- Performed statistical analysis to investigate the relationship between different global meteorological conditions (geopotential height, temperature, NAO, etc.) with Hong Kong reservoirs' yield

Scholarships and Fellowships

2020 – 2023	Future Investigators in NASA Earth and Space Science and Technology (FINESST) fellowship
2021	Brundrett Memorial EPS Fellowship The University of Texas at Austin
2016	Chung Chi College Class Scholarship The Chinese University of Hong Kong
2014, 2017	Talent Development Scholarship The Government of the HKSAR

Awards and Achievements

2022	Second Place in the American Meteorological Society Annual Meeting Oral Presentation
2021	Second Place in Student Research Symposium Oral Presentation The University of Texas at Austin
2020	Outstanding Teaching Assistant Award The University of Texas at Austin
2017	Outstanding Student Paper Award American Geophysical Union
2015, 2016	Dean's Honours List The Chinese University of Hong Kong

Teaching Experience

The University of Texas at Austin

Texas, US

Climate: Past, Present, Future

(Lead Teaching Assistant) Jan 2020 – May 2020

(Teaching Assistant) Jan 2019 – May 2019

- Coordinated among different teaching assistants
- Prepared teaching materials (PowerPoints, Experiments, and Assignments)

Physical Geology

(Teaching Assistant) Sep 2019 – Dec 2019

- Introduced different types of rocks and minerals, tectonics, and hydrology
- Hosted regular office hours to address students' questions

The Chinese University of Hong Kong

Hong Kong

Tropical Meteorology

(Teaching Assistant) Jan 2017 – May 2017

- This course was a co-teaching course for both graduate and undergraduate students
- Introduced the phenomena and motions in the tropical troposphere (e.g. ITCZ, ENSO, MJO, cyclones, and monsoons)
- Introduced methods to examine tropical atmospheric phenomena (e.g. Hovmöller diagram)

Global Environmental Change

(Teaching Assistant) Sep 2016 – Dec 2016

- Introduced the science and mechanisms behind the global warming issue
- Discussed global warming skeptics with students
- Designed materials and introduced how to analyze atmospheric data using NCL

Service, Mentoring, and Outreach

Skype a Scientist

Virtual

Skype a Scientist Program

(Scientist) Sep 2024 – Now

- Foster scientific literacy and curiosity by facilitating open Q&A and tailoring complex scientific concepts for K–12 classrooms

University of Miami

Miami, US

CIMAS Mentoring Program

(Mentor/Mentee) Sep 2024 – Jun 2024

- Provide comments and support to peer early-career scientists regarding research and professional development

The University of Texas at Austin

Texas, US

Water, Climate, and Energy seminar series

(Coordinator) Sep 2022 – May 2023

- Coordinated weekly seminars